Genes coding for “V”, “D” and “J” regions of B and T cell receptors undergo recombination

- Each receptor is made of 2 heavy and 2 light chains
- There are 2 types of light chains, λ and κ
- Variable region of Heavy chain is made of 3 segments: V, D and J
- Variable region of Light chain is made of 2 segments: V and J

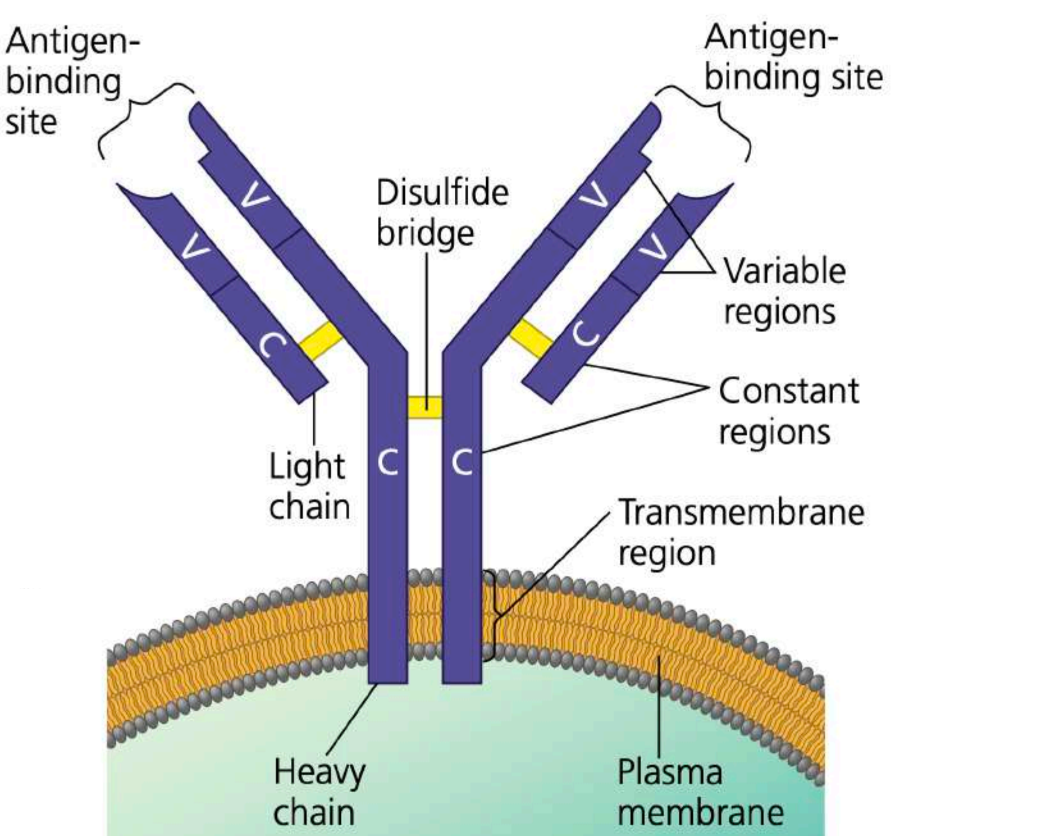
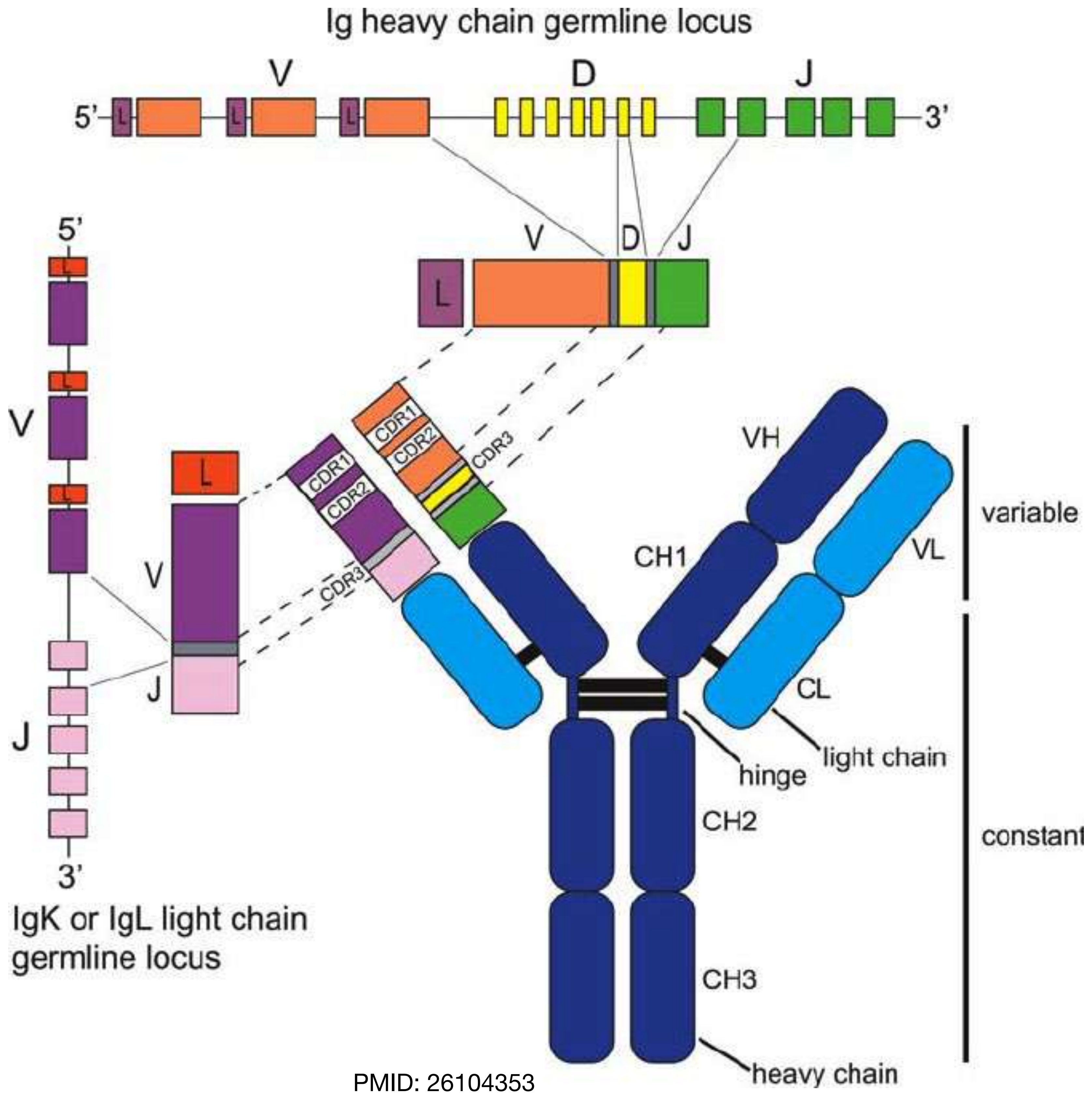


Figure 43.11 and 43.9 of Campbell’s Biology: a global approach



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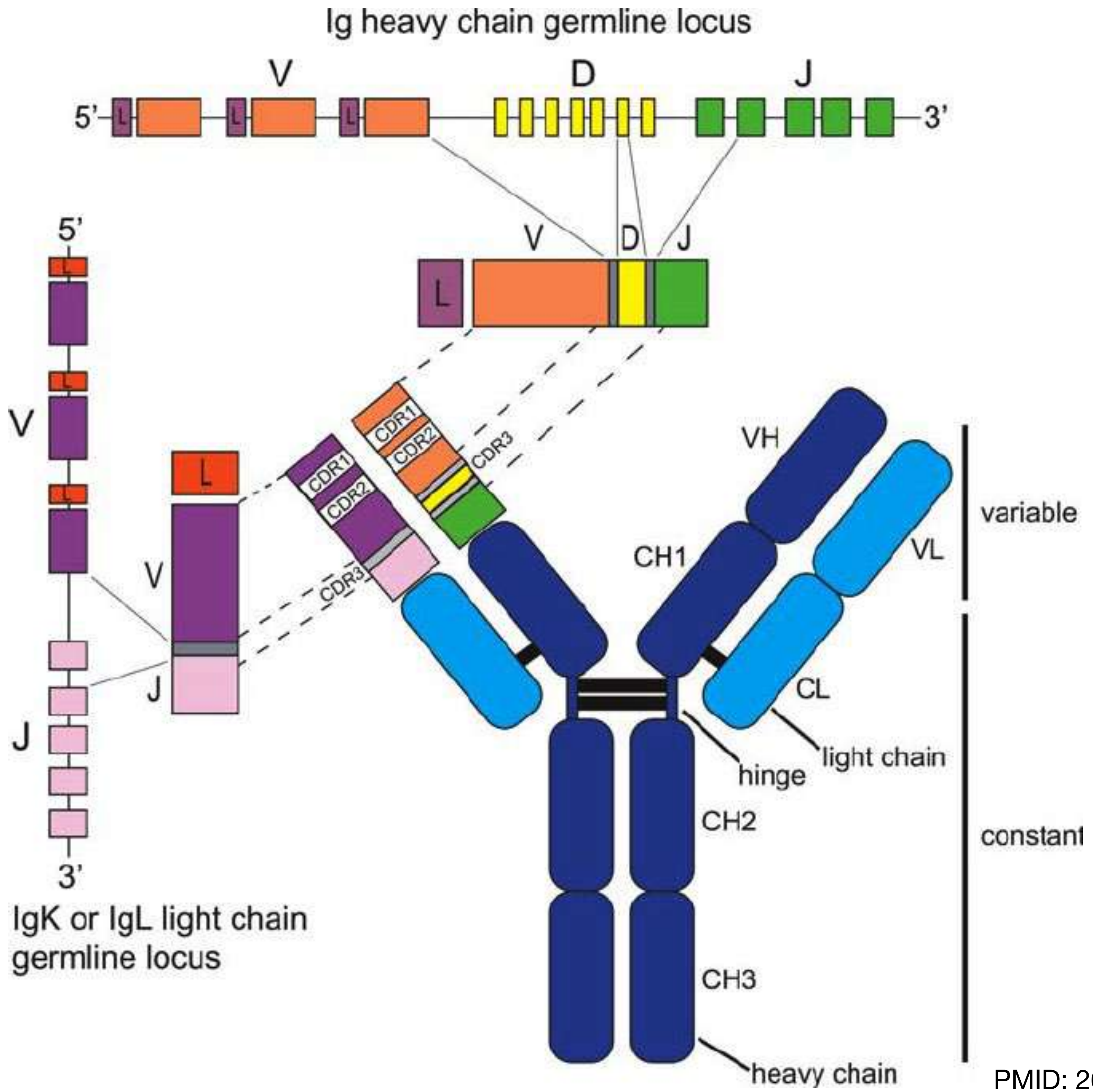
An immunoglobulin is encoded by 7 genes:
IGHV, IGHD, IGHJ, IGHC for the H chain
+
IGKV, IGKJ, IGKC for a kappa light chain

An immunoglobulin is encoded by 7 genes:
IGHV, IGHD, IGHJ, IGHC for the H chain
+
IGLV, IGLJ or IGLC for a lambda light chain

For a given receptor, the 2 light chains are identical
"D" region is only on the variable segment of heavy chain

Recombination occurs in the hematopoietic stem cells of the bone marrow - somatic recombination



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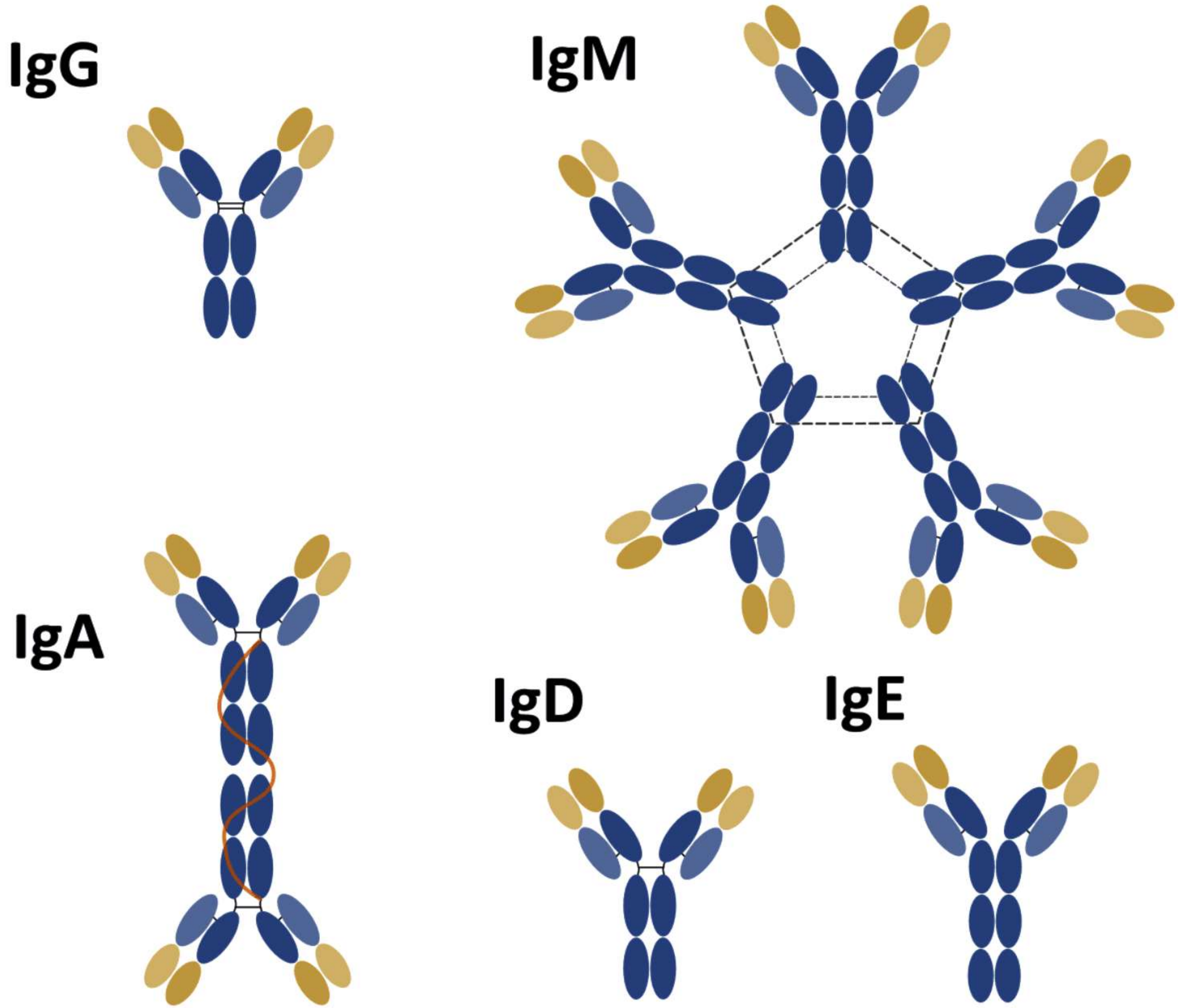
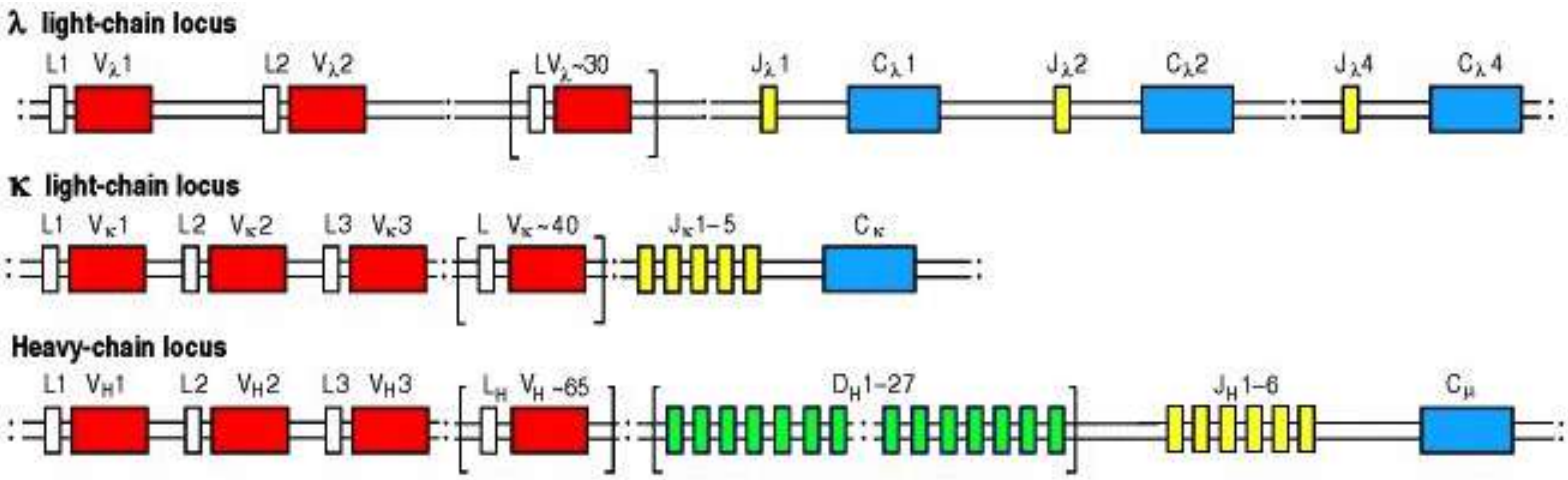
Based on heavy and light chain compositions, antibody receptors are of different immunoglobulin classes





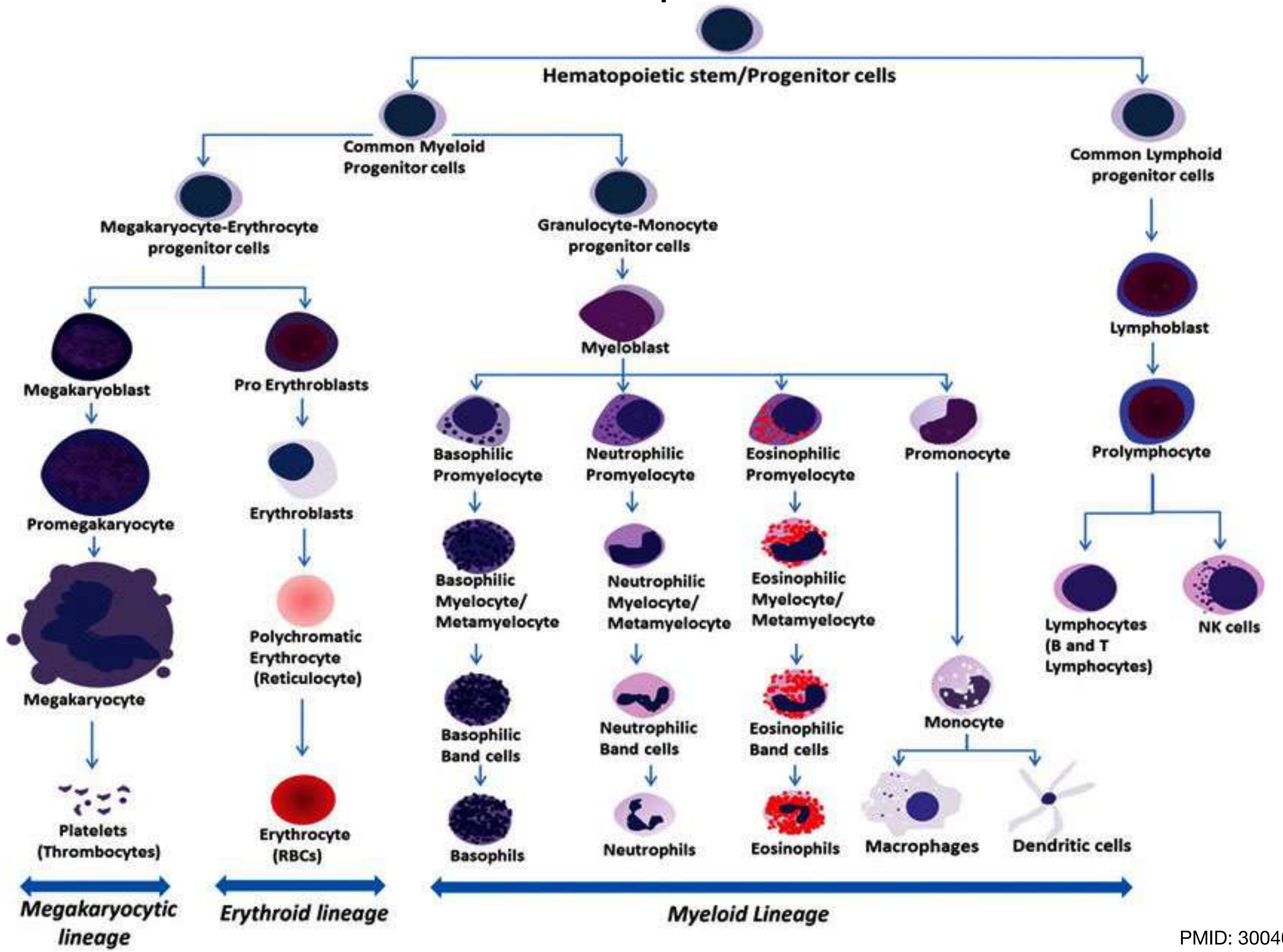
The human genome contains 176 functional immunoglobulin genes clustered in 3 loci:

- IGH on chromosome 14 (50 V, 23 D, 6 J and 9 C) - to make heavy chains
- IGK on chromosome 2 (40 V, 5 J and 1 C) - to make kappa light chains
- IGL on chromosome 22 (32 V, 5 J and 5 C) - to make lambda light chains

- During the development of B cells, the mechanisms of diversity involved in the immunoglobulin synthesis (combinatorial V-(D)-J diversity, junctional diversity and somatic hypermutations) lead to the huge potential antibody repertoire of each individual
- Estimated to comprise 10^{12} different immunoglobulins
- Limiting factor being only the number of B cells that an organism is genetically programmed to produce

https://www.ncbi.nlm.nih.gov/books/NBK27140/

Haematopoiesis



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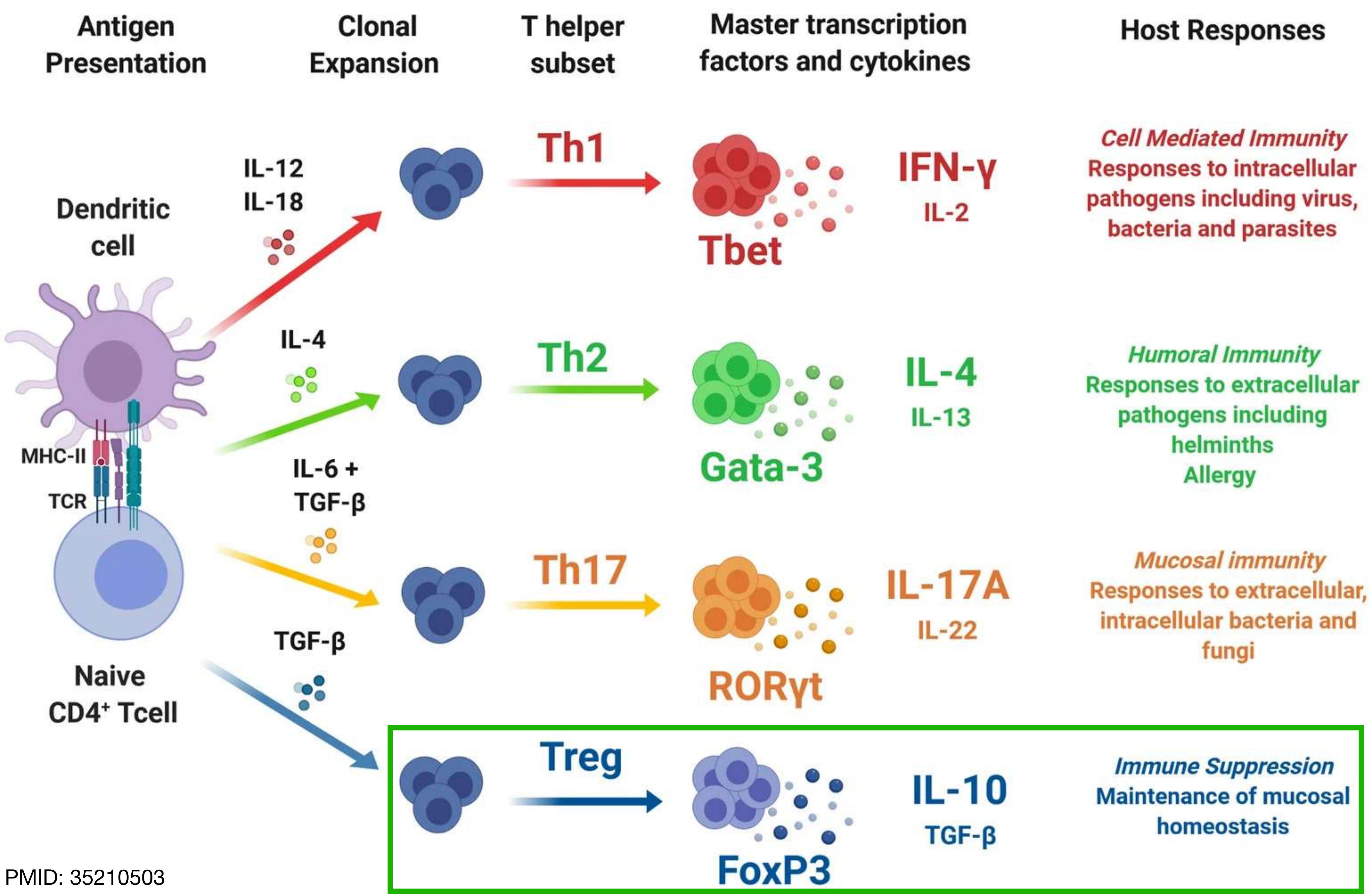
Sreelaja Nair

2025 Nobel Prize in Physiology or Medicine
Mary E. Bronkow, Fred Ramsdell and Shimon Sakaguchi



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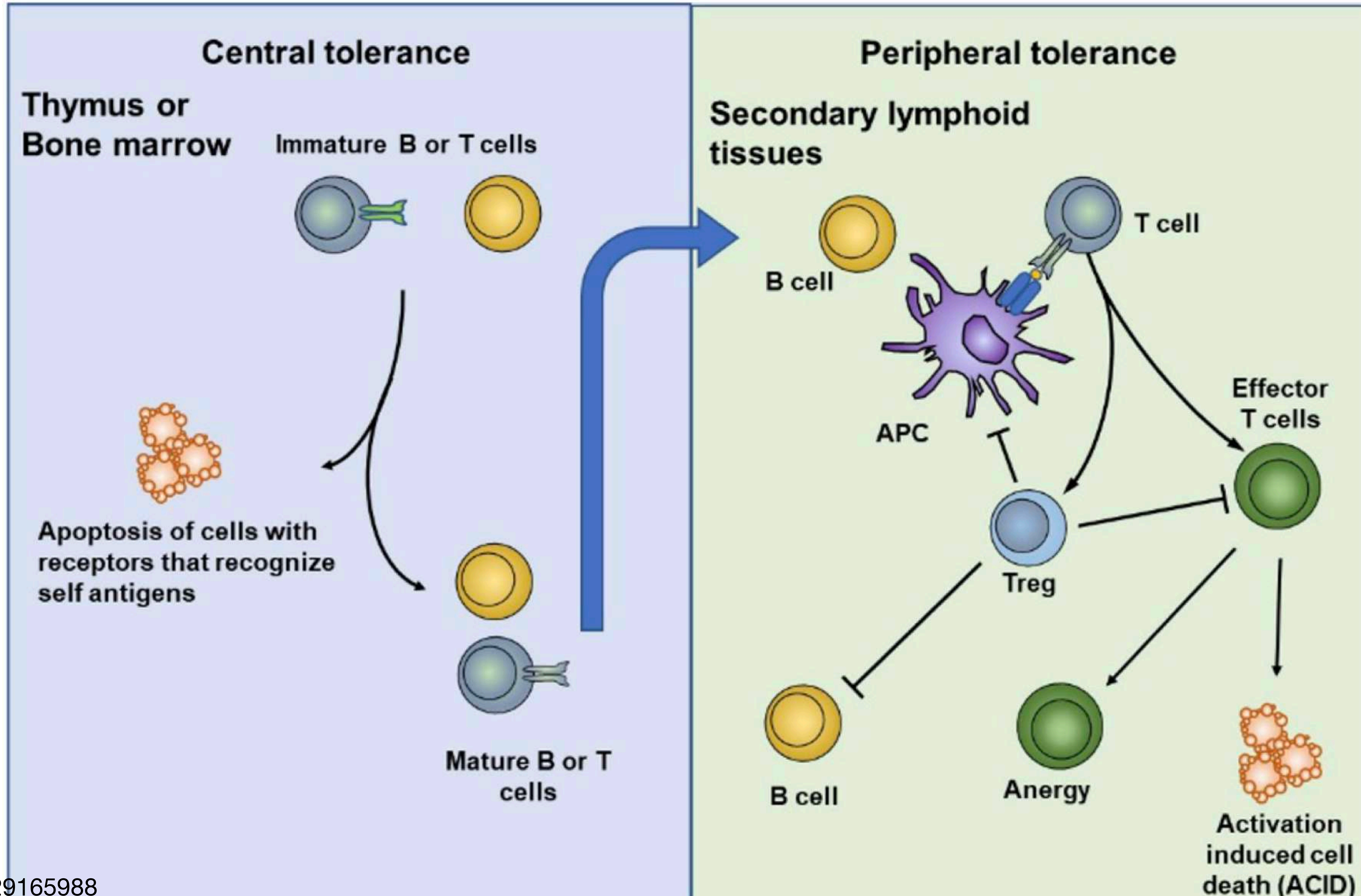
2025 Nobel Prize in Physiology or Medicine
Mary E. Bronkow, Fred Ramsdell and Shimon Sakaguchi

Discovery of Tregs and peripheral tolerance
Now a key to developing therapies for cancer and autoimmune diseases



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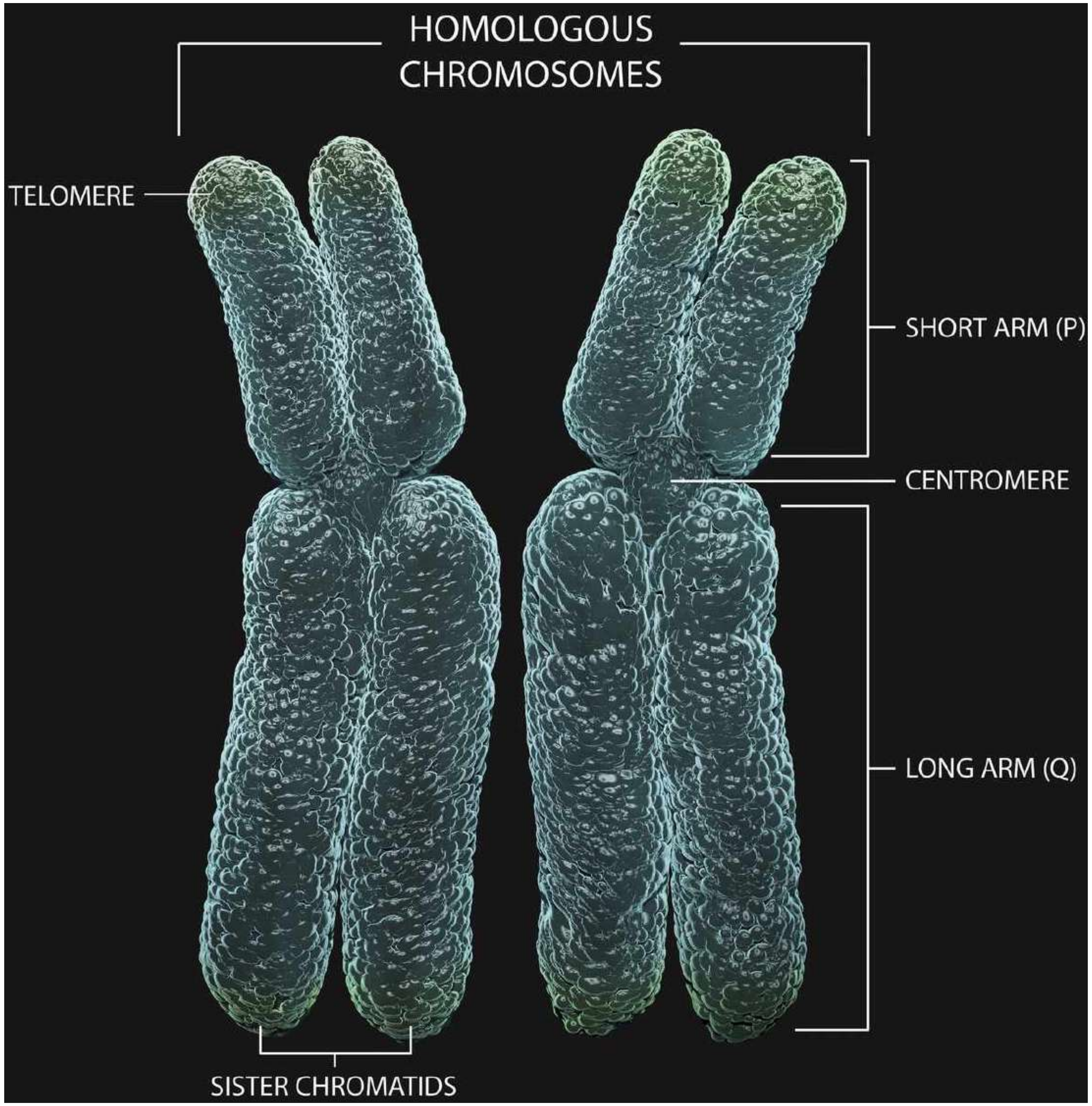
DNA recombination

- Involves the exchange of genetic material
- Can occur between different regions of the same chromosome (example: generation of antibody diversity)

- Can also occur between multiple chromosomes
- Exchange of DNA segments results in formation of “junctions” known as “Holliday Junction”
- These junctions need to be resolved to separate the individual DNA segments
- Type of resolution of Holliday Junction results in different types of of end results for the exchange event

Chromosomes

- A pair of same chromosomes = homologous chromosomes
- One half of one chromosome = sister chromatid
- Sister chromatids come into existence in S-phase of mitosis



What does mitosis separate - Chromatids or Chromosomes?

What does meiosis separate - Chromatids or Chromosomes?

What does the mature sperm contain - Chromatids or Chromosomes?

What does the mature egg contain - Chromatids or Chromosomes?

What does the one-cell embryo contain - Chromatids or Chromosomes?

What does the interphase cell contain - Chromatids or Chromosomes?

DNA recombination

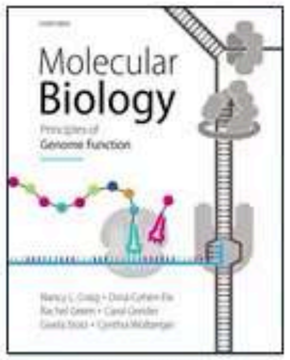
- Involves the exchange of genetic material
- Can occur between different regions of the same chromosome (example: generation of antibody diversity)

- Can also occur between multiple chromosomes
- Recombination during meiosis non-sister (homologous) chromosomes result in genetic diversity
- Recombination during mitosis between sister chromatids - usually used to repair DNA (restore lost information due to damage)

- Exchange of DNA segments results in formation of “junctions” known as “Holliday Junction”
- These junctions need to be resolved to separate the individual DNA segments
- Type of resolution of Holliday Junction results in different types of of end results for the exchange event

DNA recombination and junction resolution

- During meiosis, recombination (shown below) occurs during Prophase of meiosis I
- During mitosis, recombination occurs during S or G2 phase of cell cycle



Molecular Biology: Principles of Genome Function

Second Edition

OXFORD UNIVERSITY PRESS

Animation 14:

Holliday junction resolution

Animation produced by Connor Hendrich

https://www.youtube.com/watch?v=MvnWxN81Qps